

Integrating LLM-Based Nutrient Estimation from Chinese Diet Logs for Better Continuous Glucose Forecasting

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Abstract—Accurate short-term forecasting of blood glucose is essential for safe diabetes self-management, yet existing forecasters seldom leverage nutrients estimated from free-text meal logs. This estimating nutrient is especially challenging for Chinese meals due to mixed dishes and diverse preparation styles. In this study, we investigate whether large language model (LLM)-based nutrient estimation from Chinese diet logs can improve dynamic glucose forecasting. Five Chinese-capable LLMs for Chinese texts (GPT5-mini, DeepSeekv3.1, Llama3.1-70B, GLM4.5, Qwen3-235B) are prompted to estimate macronutrients and weighted GI for meals. We also compare two pipelines for nutrition estimation: database-first with LLM fallback and LLM-only with retrieval-augmented generation (RAG). We propose a seasonal-patch Transformer architecture designed for multi-source feature fusion, integrating dynamic sequences, static patient data, diet word embeddings, and the LLM-estimated nutritional profiles. Experiments on 112 real-world Chinese patients (ShanghaiT1DM $n = 12$, ShanghaiT2DM $n = 100$) showed that, versus a no-LLM baseline, LLM-augmented models reduced 3-h postprandial RMSE by up to 1.2 mg/dL (36.06 $\rightarrow$ 34.86 mg/dL) and achieved the best full-period RMSE of 26.66 mg/dL, comparable to a recent large pretraining forecasting model. Forecast gain correlates strongest with carbohydrate ($p = 0.94$) and fat ($p = 0.77$) estimation accuracy. These results demonstrate that LLM-based nutritional feature engineering provides a critical pathway toward more precise, personalized CGM forecasting and the realization of clinical digital twins.

Keywords—Continuous Glucose Monitoring (CGM), Glucose Forecasting, Large Language Models (LLM), Nutrient Estimation, Diabetes Management

I. INTRODUCTION

Continuous glucose monitoring (CGM) is widely utilized in the self-management of type 1 and type 2 diabetes (T1D and T2D), and has been shown to improve clinical outcomes [1]. CGM measures interstitial glucose levels in real time through minimally invasive subcutaneous methods. However, there is some time delay in the CGM measurement values [2], making it necessary to make forecasting of continuous glucose.

Meanwhile, accurate forecasting of blood glucose levels is critical for diabetes self-management, enabling timely insulin adjustment, hypoglycemia/hyperglycemia avoidance, and ultimately improving health outcomes [3].

In continuous glucose forecasting, prior work spans data-driven, physiology-based, and hybrid approaches. A review of 138 models shows that deep learning models (e.g., transformer, LSTM) have been widely adopted, and some works incorporate exogenous signals (activity, meals) [4]. Continuous glucose forecasting relies on past glucose trajectories and may add exogenous features such as insulin dosage, physical activity, or diet information. The inclusion of such exogenous features has been demonstrated to improve the accuracy of blood glucose forecasting [5]. Among exogenous features, carbohydrate intake is the most frequently used feature in predictive models, given that carbohydrate is metabolized into blood glucose [6].

However, few studies incorporated full macronutrients into glucose forecasting. This phenomenon may be attributable to limitations in existing datasets. For instance, the widely used Ohio-T1D dataset only includes carbohydrate intake data [7]. The underlying reason for this is that dietary records collected in free-living settings are often unstructured and the high cost to make detailed nutrient breakdowns [8]. While established nutrition databases are available for user access and recording, the complexity and variability of mixed dishes, such as Chinese cuisine, pose a significant challenge to accurate nutritional assessment using these resources [9]. A recent work applied Bayesian networks for predicting 2-hour postprandial blood glucose using ShanghaiT2DM data. The nutritional information was introduced by expert knowledge instead of automated tools [10].

Large language models (LLMs) are increasingly utilized in medicine and nutrition. Their capacity to generate and annotate data has facilitated the development of enhanced clinical forecasting models [11], [12]. Furthermore, LLMs offer the capability to process natural language dietary descriptions and generate structured nutrient estimations. Recent works have evaluated LLMs' capacity to estimate macronutrients from meal descriptions. LLMs can provide competitive and significantly faster estimates than professional nutritionists [13]. For blood glucose forecasting,

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LLM-based nutrition estimation has been shown to enhance accuracy in virtual diabetes patients' glucose forecasting [14]. This study hypothesizes that LLMs will yield comparable benefits when forecasting real-world CGM datasets.

In this work, we integrate LLM-estimated nutrient features into a transformer-based forecasting model and evaluate its performance on two public real-world Chinese diabetes cohorts (ShanghaiT1DM, ShanghaiT2DM). The contributions are:

- We propose a framework to fuse LLM-based macronutrients and glycemic index (GI) estimation with static and dynamic clinical features for glucose forecasting.
- We systematically compare 5 leading LLMs as estimators and ablate their contributions.
- We perform correlation analyses linking forecasting error improvements with estimation accuracy relative to nutritional databases.

The rest of the paper is organized as follows. Section II reviews related work. Section III details our data preprocessing, LLM estimation pipeline, and forecasting architecture. Section IV presents experimental design and key results. Section V discusses implications, limitations, and future work.

II. METHOD

In this section, we detail the data preprocessing, nutrition estimation pipeline, the forecasting task definition, forecasting model architecture, and evaluation protocols.

A. Dataset and Preprocessing

We use the publicly described ShanghaiT1DM ($n = 12$) and ShanghaiT2DM ($n = 100$) datasets, collected under real-life conditions with CGM sampling by flash glucose monitoring devices (FreeStyle Libre H, Abbott Diabetes Care, Witney, UK) every 15 minutes over 3–14 days per patient [8]. Each patient record includes static demographic and clinical features (e.g., age, sex, BMI, HbA1c, lab measures, presence of complications), medication logs, and daily dietary logs in Chinese text.

Preprocessing steps are as follows:

Static features: Patient type is labeled based on diabetes classification. Comorbidities and complications are represented as a count of unique disease types. Categorical features were one-hot encoded. Missing values are imputed using feature-wise means (for continuous features) or modes (for categorical features).

Dynamic features: The dynamic features include medication information aligned with CGM, macro-nutrient estimation data, diet log text, and recording time. For any continuous glucose monitoring (CGM) data gaps of ≤ 1 hour (maximum four 15-minute intervals), linear interpolation is applied to impute missing values. The missing values of medication dosages are regarded as 0. For dietary logs, 'unrecorded' entries are treated as missing data and imputed with the mean CGM value for the corresponding time interval; otherwise, no food intake was assumed (value of 0). Regular expressions are employed to parse Chinese food descriptions, extracting quantity, units, and food names. The food names were then deduplicated and manually checked for incorrect

spellings. We use LLMs to estimate the nutritional content of each 100g (or ml) of food, and then sum up the results weighted by the consumption amount. These food names are then converted to word embeddings (hereafter referred to as diet embedding) using the pre-trained Chinese language model BGE-M3[15][16]. The recording time of CGM has been simplified to 0-24 hours.

Normalization: Continuous features (static or dynamic) are standardized (z-score) based on training set statistics.

B. LLM-based Nutrition Estimation

Chinese original version:

你是一位资深营养师。请根据你的专业知识，估算菜肴 {food_name} 每100g或ml的营养成分，标准参考中国食物成分表标准版（第6版）以及其他可能有帮助的资料。如果菜肴 {food_name} 中有多种菜肴，请按照一般食用比例估算营养成分。如果你认为菜肴名称无法精确估算，请使用近似估算，不允许输出null。如果菜肴 {food_name} 中存在非100g或ml的食用量和单位，则直接输出菜肴 {food_name} 的营养成分，不要做任何单位标准化，并在json中包含食用量和单位。请严格按照下面的JSON格式返回结果，不要包含任何额外的文本。

要求的营养素包括：

- 热量 (kcal)
- 蛋白质 (g)
- 脂肪 (g)
- 碳水化合物 (g)
- 膳食纤维 (g)
- 加权GI

JSON输出格式示例：

```
{
  "thinking_process_step_by_step": "这里描述你是如何估算这些营养成分的，步骤清晰。",
  "calories": 150.0,
  "protein": 10.5,
  "fat": 8.2,
  "carbohydrates": 5.1,
  "fiber": 2.3,
  "gi": 3.5,
  "食用量和单位（如果有）": "100g或ml"
}
```

English translation version:

You are a Senior Nutritionist. Based on your professional knowledge, and referencing the 6th edition of the Chinese Food Composition Table (CFC Standard) and any other helpful resources, estimate the nutritional composition per 100 grams/milliliters of the dish: {food_name}.

If {food_name} comprises multiple dishes, estimate the nutritional content based on their typical consumption ratio.

If you believe the dish name is insufficient for a precise estimation, please use the closest possible approximation.

Outputting 'null' is strictly prohibited.

If the serving size and unit for the dish {food_name} are not 100g/ml, output the nutritional composition for the specified serving size and unit directly, without any unit standardization, and include the serving size and unit in the JSON output.

You must strictly adhere to the following JSON format for the result, excluding any additional text.

The required nutrients include: Calories (kcal), Protein (g), Fat (g), Carbohydrates (g), Dietary Fiber (g), and Weighted Glycemic Index (GI).

JSON Output Format Example

```
{
  "thinking_process_step_by_step": "Describe the steps you took to estimate the nutritional content here, clearly and sequentially.",
  "calories": 150.0,
  "protein": 10.5,
  "fat": 8.2,
  "carbohydrates": 5.1,
  "fiber": 2.3,
  "gi": 3.5,
  "Consumption amount and unit (if any)": "100g or ml"
}
```

Fig. 1. The CoT-like prompt template for LLMs when directly estimating, with '{food_name}' as the placeholder.

We evaluate three LLM-based nutrient estimation strategies: direct estimation, a hybrid LLM-database approach that only estimates components not covered by the database, and an exploratory novel RAG solution.

Five large language models (LLMs) are employed to process the Chinese dietary logs. These included the closed-source model GPT-5-mini-2025-08-07 [17], along with the open-weight models Deepseek-v3.1-2025-08-21 [18], Llama-3.1-70B [19], glm-4.5-air [20], and Qwen3-235b-a22b-fp8 [21]. The selection of these models is driven by task characteristics and linguistic considerations. Given the short context and batch-processing demands of nutrient labeling, we prioritized smaller, more efficient models over their larger or thinking-enabled counterparts. Additionally, for this Chinese language task, we favor models developed within China, leveraging their inherent advantages in processing and understanding Chinese text.

We use a CoT-like prompt strategy inspired by the NutriBench research [14] (see Fig. 1). The LLMs are instructed to adopt the persona of a professional nutritionist and to output nutritional information in a structured JSON format. This format included the following values per 100 g (or ml) of food: Calories (kcal), Protein (g), Fat (g), Carbohydrate (g), Dietary Fiber (g), and Weighted Glycemic Index (GI). The JSON structure's initial key is designated as "thinking process step-by-step". Meanwhile, the prompt instructs the LLMs to record whether non-standard units are present in the food name, as using only regular expressions for unstructured diet logs may still lead to exceptional cases (for example, recording "a slice of bread" instead of "bread 100g"). Recording units helps improve the accuracy of calculating total meal calories throughout the automated process.

We test the performance of a hybrid LLM-database strategy on the top 3 LLMs, prioritizing database exact matches for nutritional information and employing LLM estimates for unmatched data and GI. This approach aligns with the computational resource optimization required for real-world deployment. The nutrition database is constructed from the 6th edition of the Chinese Food Composition Table (CFC Standard) and the Chinese food dish nutrition database collected from the Internet (provided by Health Hope Technology Co., Ltd., Beijing, China).

Retrieval-augmented generation (RAG) techniques are not initially prioritized in this study, based on previous findings indicating that RAG can introduce noise into nutrition estimation when used with LLMs [14]. However, we explore a novel RAG pipeline based on Deepseekv3.1. This RAG pipeline first prompts the model to check whether the retrieved results are sufficient. This retrieval is based on the word segmentation function of the jieba package. If not, the "food" is decomposed into its constituent components, and then fuzzy matching is performed of these ingredients against the CFC database. The prompt used for ingredient decomposition is shown in Fig. 2. The top-5 retrieved results for each ingredient are then provided to the LLM for nutrition estimation. These results were appended to the CoT-like prompt in the following format: “基础食材营养信息如下（均为每100克）：{RAG_result}。如果我提供的食材，对于估算【{food_name}】没有参考意义，请根据你的知识库对【{food_name}】进行直接估算。” (English translation: “The basic nutritional information of the ingredients is as follows (all per 100 grams) : {RAG_result}.

If the ingredients I provide have no reference value for estimating [{food_name}], please make a direct estimate of [{food_name}] based on your knowledge base.”).

Chinese original version:

你是一名营养数据分析师。我需要处理以下食品'{food_name}'。

1. 我在数据库中找到了以下相关项：{fuzzy_names if fuzzy_names else "None"}。

2. **决策与行动**:

- 如果您认为所提供的食材足以对'{food_name}'做出合理的营养评估，将'sufficient'设置为true。
- 如果这些食材不充足，则将'sufficient'设置为false，并根据《中国食物成分表》提供'{food_name}'所需的标准食材清单。
- 不常见的食品别名请改为常用名
- 仅包含人类可食用的食材，且是制作合理营养评估所必需的。香料、盐、水不应包括在内。

3. **JSON输出**：仅以JSON格式回复。

- 成功时：{"sufficient": true, "ingredients": []}
- 失败/故障时：{"sufficient": false, "ingredients": ["ingredient1", "ingredient2", ...]}

English translation version:

You are a nutritional data analyst. I need to process the following food item: '{food_name}'.

1. I found the following items in the database: {fuzzy_names if fuzzy_names else "None"}.

2. **Decisions and actions**:

- if you think items provided enough to '{food_name}' make reasonable nutrition assessment, will be 'sufficient' is set to true.
- If these items are not enough, they will be 'sufficient' set to false, and according to the "food ingredients in China" to provide '{food_name}' required standard ingredient list.
- Unusual food alias please instead of common name contains only human edible ingredients, and is used for making reasonable nutrition assessment required.
- Spices, salt and water should not be included.

3. **JSON Output** : Reply in JSON format only

- On success: {"sufficient": true, "ingredients": []}
- On failure/failure: {"sufficient": false, "ingredients": ["ingredient1", "ingredient2", ...]}

Fig. 2. The prompt template to decompose food ingredients

C. Forecasting Task Definition

We consider the task of forecasting CGM measurements in patients using a multi-source feature fusion time series framework. Let B denote the batch size, L the sequence length, F_d the number of dynamic features, and F_s the number of static features. The objective is to predict the target CGM values at a future horizon of H steps.

The input integrates both dynamic and static features:

- Dynamic Features $\mathbf{X}_{\text{dyn}} \in \mathbb{R}^{B \times L \times F_d}$** , could including: CGM measurements, medication records, estimated nutrition intake, diet word embeddings and recording times.
- Static Features $\mathbf{X}_{\text{stat}} \in \mathbb{R}^{B \times F_s}$** capturing patient-specific attributes such as age, BMI, and baseline clinical measurements.
- Dynamic Missingness Mask $\mathbf{M}_{\text{dyn}} \in \{0, 1\}^{B \times L \times F_d}$** indicating observed (1) and missing (0) entries in the dynamic features.
- Static Missingness Mask $\mathbf{M}_{\text{stat}} \in \mathbb{R}^{B \times F_s}$** indicating observed (1) and missing (0) entries in the Static features.

The forecasting target is the future CGM measurements:

$$\mathbf{y} = [\text{CGM}_{t+1}, \dots, \text{CGM}_{t+H}] \in \mathbb{R}^{B \times H}. \quad (1)$$

Given the past dynamic sequence $\mathbf{X}_{\text{dyn}}$ and static features $\mathbf{X}_{\text{stat}}$, the goal is to learn a mapping $f: (\mathbf{X}_{\text{dyn}}, \mathbf{X}_{\text{stat}}) \mapsto \mathbf{y}$, that predicts future CGM values over the forecasting horizon H .

The model is trained to minimize the mean squared error (MSE) between predicted and true CGM measurements:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{\sum_{i,h} \mathbf{M}_{i,h}^{\text{CGM}}} \sum_{i=1}^B \sum_{h=1}^H \mathbf{M}_{i,h}^{\text{CGM}} (\hat{y}_{i,h} - y_{i,h})^2, \quad (2)$$

where $\mathbf{M}^{\text{CGM}} \in \{0,1\}^{B \times H}$ indicates missingness mask of CGM points.

D. Forecasting Model

The Transformer-based forecasting model includes seasonal decomposition [22], patched transformer modules [23], and multiple feature branches (static, dynamic, diet). The architecture also allows for equitable comparison of different feature configurations, specifically with and without static features, diet embeddings, and nutrition estimation (Fig. 3).

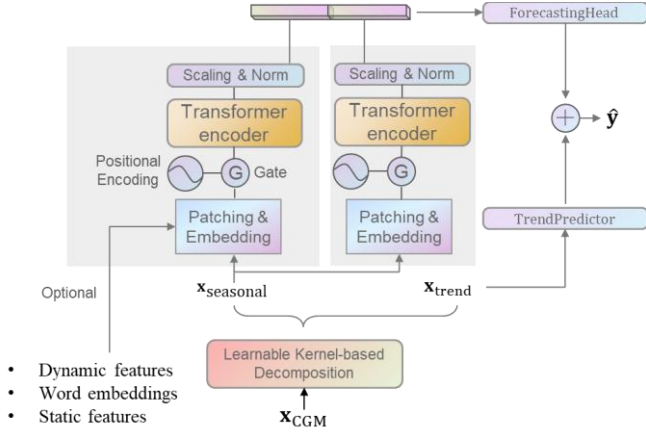


Fig. 3. Architecture of the proposed forecast model

Below is a description of each major component:

1) Decomposition and Branching

The CGM time series ($\mathbf{x}_{\text{CGM}}$) is first decomposed into seasonal ($\mathbf{x}_{\text{seasonal}}$) and trend ($\mathbf{x}_{\text{trend}}$) components via a learnable kernel-based operator:

$$\mathbf{x}_{\text{CGM}} = \mathbf{x}_{\text{seasonal}} + \mathbf{x}_{\text{trend}}. \quad (3)$$

The seasonal component is then processed by two parallel Transformer encoders: the **Main Branch** for multi-modal feature fusion and the **Side Branch** focusing solely on the target's seasonal history. This parallel design enables a direct comparison of various feature settings.

2) Main Branch: Multi-source Feature fusion

The seasonal data and all dynamic features ($\mathbf{X}_{\text{dyn}}$) are transformed into $N_p = L/P$ patches non-overlapping patches of length P . The main branch patch embedding ($\mathbf{E}_{\text{patch}}$) integrates all external information through controlled, learnable gating mechanisms:

- Dynamic features ($\mathbf{X}_{\text{dyn}}$) and missing masks ($\mathbf{M}_{\text{dyn}}$) are embedded and reshaped. The $\mathbf{x}_{\text{seasonal}}$ is embedded as $\mathbf{E}_{\text{seasonal}}$.

- Diet embeddings ($\mathbf{E}_{\text{diet}}$) are reshaped.
- Static features ($\mathbf{X}_{\text{stat}}$) and missing masks ($\mathbf{M}_{\text{stat}}$) are embedded and broadcast across all patches.

This fusion process is formalized as an additive operation with respective learnable patch gates ($\mathbf{G}$):

$$\mathbf{E}_{\text{patch}} \leftarrow \mathbf{E}_{\text{seasonal}} + \mathbf{G}_{\text{dyn}} \cdot \mathbf{E}_{\text{aux}} + \mathbf{G}_{\text{diet}} \cdot \mathbf{E}_{\text{diet}} + \mathbf{G}_{\text{stat}} \cdot \mathbf{E}_{\text{stat_patch}}. \quad (4)$$

After positional encoding (PE) and layer normalization, the Transformer encoder produces $\mathbf{H}_{\text{main}}$. The main representation $\mathbf{R}_{\text{main}}$ is obtained via mean pooling and layer normalization with learnable scaling (γ_{dyn}):

$$\mathbf{H}_{\text{main}} = \text{Transformer}_{\text{main}}(\text{PE}(\mathbf{E}_{\text{patch}})), \quad (5)$$

$$\mathbf{R}_{\text{main}} = \text{LayerNorm}(\gamma_{\text{dyn}} \cdot \text{mean}(\mathbf{H}_{\text{main}})). \quad (6)$$

3) Side Branch: Seasonal Representation Learning

The Side Branch encodes the seasonal component exclusively. The sequence $\mathbf{x}_{\text{seasonal}}$ is divided into $\mathbf{P}_{\text{side}}$.

$$\mathbf{H}_{\text{side}} = \text{Transformer}_{\text{side}}(\text{PE}(\mathbf{P}_{\text{side}})) \quad (7)$$

$$\mathbf{R}_{\text{side}} = \text{LayerNorm}(\gamma_{\text{dyn}} \cdot \text{mean}(\mathbf{H}_{\text{side}})) \quad (8)$$

4) Representation and Forecasting

Both Main and Side Transformer encoders generate patch-level representations, which are pooled and combined into final representations ($\mathbf{R}_{\text{main}}$ and $\mathbf{R}_{\text{side}}$). These representations are fused and passed to the forecasting head:

$$\mathbf{R}_{\text{fused}} = \text{FusionProj}([\mathbf{R}_{\text{main}}; \mathbf{R}_{\text{side}}]) \quad (9)$$

The final forecast ($\hat{\mathbf{y}}$) is the summation of the predicted seasonal component ($\hat{\mathbf{y}}_{\text{seasonal}}$) from the $\mathbf{R}_{\text{fused}}$ and the linearly predicted trend component ($\hat{\mathbf{y}}_{\text{trend}}$) from the $\mathbf{x}_{\text{trend}}$:

$$\hat{\mathbf{y}} = \text{ForecastingHead}(\mathbf{R}_{\text{fused}}) + \text{TrendPredictor}(\mathbf{x}_{\text{trend}}) \quad (10)$$

E. Evaluation

We evaluate two forecasting tasks: full-period glucose forecasting and postprandial glucose forecasting. For full-period forecasting, we use a rolling-window scheme. For postprandial glucose forecasting, we select data segments where: (1) a meal event occurred within the observation window, and (2) the forecasting window started within 30 minutes (2 data points) following the last meal event. The evaluation metric is root mean square error (RMSE) between predicted and true glucose values over the 12-point horizon, excluding missing CGM points.

In terms of the input features of the forecasting model, the options of whether to use static features, LLM for annotating nutritional features, and diet embeddings are controlled. The baseline model uses only historical glucose data. The historical CGM and dynamic medication information are used in all the other experiments.

Additionally, we conduct a comprehensive analysis of the nutrient estimation results, including calculations of RMSE and mean absolute error (MAE) against database values, as well as correlation analyses. This database lacks GI data, so only macronutrients are included in the evaluation. We calculate the Spearman correlation coefficient between the RMSE of the LLMs' nutrient estimates (compared to the

database) and the average RMSE of the LLMs' blood glucose forecasting. This aims to uncover the relationship between the accuracy of nutrient estimation and the capability for blood glucose prediction.

III. RESULT

A. Experiment Settings

The dataset is split into patient-level training, validation, and test sets (7:1:2 ratio) with stratification by patient type (T1D, T2D). Data splitting employs a random seed of 42 for reproducibility. For each patient, we slide a 48-point observation window and predict the next 12 points (3 hours). The full-period glucose forecasting task consisted of 120,810 input-output pairs, while the postprandial glucose forecasting task included 6,631 pairs.

The temperature value of LLMs is set to 1 as recommended for most tasks. There are a total of 1944 types of foods that can be directly estimated using LLMs, among which 91 types can be precisely matched in the CFC Standard database and 384 types can be precisely matched in the dish nutrition database.

The forecasting models training is conducted using the Muon optimizer with Adam [24]. Muon is applied to all parameters with dimensions greater than or equal to 2 (with a decay rate of 0.1, a peak learning rate of 0.002) because it is designed specifically for matrix multiplication, while Adam is used for parameters with dimensions less than 2 (with a decay rate of 0.1, a peak learning rate of 0.0001).

The hidden layer dimension of the forecasting model is 256. Each Transformer architecture inside consists of 6 encoder layers with 8 attention heads. We train the forecasting model with a batch size of 2048 for a maximum of 200 epochs, employing a 25-epoch early stopping strategy. The model is developed by Python 3.12 and Torch and trained on the NVIDIA V100 GPU.

B. Forecasting results

As described in the Methods section, our experiments are conducted in multiple stages. But for clarity and completeness, all experimental configurations are summarized in Fig. 4.

For full-period forecasting, the baseline model using only historical glucose data achieved the highest RMSE. Among the LLMs tested, DeepSeekv3.1_db achieved the optimal overall performance with an RMSE of 26.66 mg/dL. This result represents a small improvement over the best-performing w/o model (RMSE of 26.98 mg/dL).

For postprandial CGM forecasting, the inclusion of LLM-estimated nutrients led to more noticeable improvements. The DeepSeekv3.1_db method, utilizing only static features, achieved the best performance (RMSE = 34.86 mg/dL), representing an improvement of approximately 1.2 mg/dL compared to the best model without LLM-estimated nutrients (RMSE = 36.06 mg/dL). Models based on DeepSeekv3.1 and gpt5-mini demonstrated significantly better average RMSE scores, while GLM4.5, Llama3.1, and Qwen3 performed poorly.

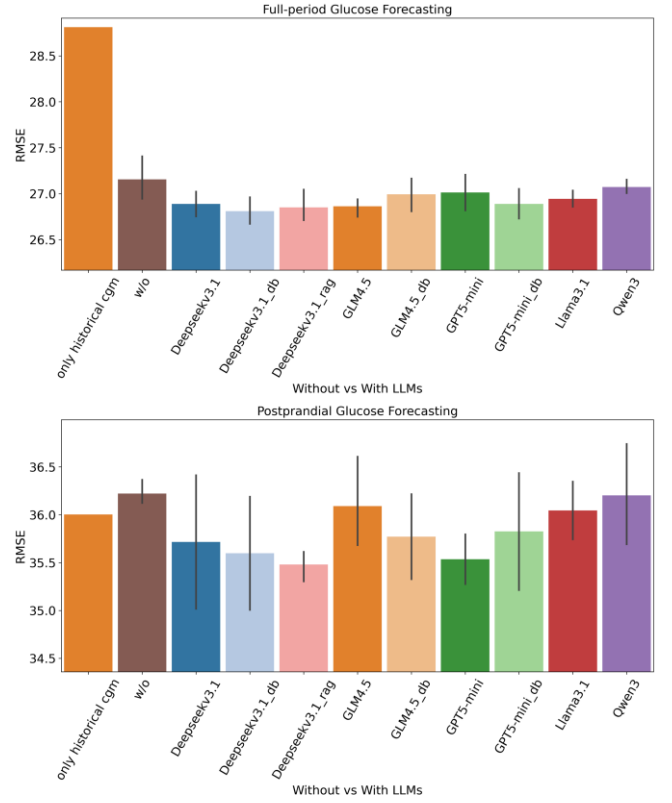


Fig. 4. Average forecasting performance of different LLMs across different feature settings (with/without static features/diet embeddings). The w/o group is without LLM. The “db” suffix indicates the method of combining with the nutrition database, and the “rag” suffix indicates that the RAG technique is used.

Incorporating external databases sometimes yielded further improvement. For example, DeepSeekv3.1 + database achieved better postprandial forecasting when using only static features (RMSE = 34.86 mg/dL) than the counterpart without a database, whose RMSE is 36.06 mg/dL. However, including the database was not consistent. Incorporating the database did not improve GPT5-mini's postprandial glucose mean forecasting performance.

The RAG method improved glucose level forecasting and exhibited greater robustness to the incorporation of other features, especially for postprandial CGM forecasting. However, the optimal result is not derived from the method.

Fig. 5 shows the mean RMSE across feature combinations, highlighting the interaction between static features and diet word embeddings under LLM and non-LLM settings. LLM-estimated macronutrients led to lower RMSEs. However, the LLM-induced improvement decreased after injecting diet embeddings, indicating that the word embeddings may have introduced excessive noise rather than useful information. When both static and diet word embeddings were simultaneously included, the overall performance of LLM-augmented models became slightly worse than that of their non-LLM counterparts.

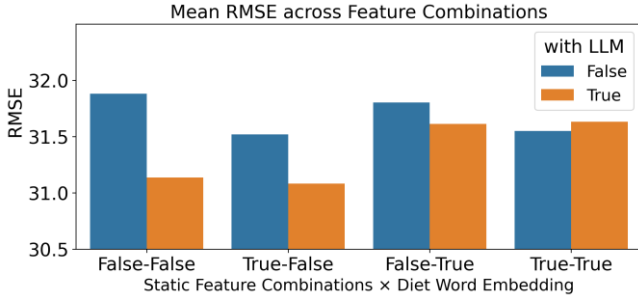


Fig. 5. Average forecasting performance of different feature settings (with/without static features/diet embeddings).

C. Nutrition Estimation analyses

Considering only the strategy of direct nutrient estimation, the LLMs were ranked based on their overall and postprandial blood glucose forecasting performance. GPT5-mini exhibited the highest accuracy, followed by Deepseekv3.1, GLM4.5, Llama3.1, and Qwen3 (TABLE 1). When incorporating the database and RAG, Deepseekv3.1_rag achieved the highest average performance, closely followed by Deepseekv3.1_db.

Although LLM-based nutrient estimations improve CGM forecasting, they still exhibit biases compared to database values. These biases were quantified (TABLE 1) using precisely matched food items from our mixed database, which included both raw ingredients and composite dishes. Deepseekv3.1, GLM4.5, and GPT5-mini each had 475 matches, while Llama3.1 and Qwen3 showed slightly lower counts of 458 and 453, respectively, due to some outputs lacking unit annotations that were consequently excluded. Notably, the RAG’s 384 entries consisted entirely of items not found in the CFC Standard database for fair comparison. Correlation analysis between the biases and forecasting rankings revealed the following order of association: carbohydrates ($\rho = 0.94$), calories ($\rho = 0.94$), fat ($\rho = 0.77$), fiber ($\rho = 0.6$), and protein ($\rho = -0.14$).

TABLE I. PERFORMANCE COMPARISON OF DIFFERENT LLMs ACROSS GLUCOSE AND NUTRITION METRICS

Rank Metric	DeepSeek v3.1_rag	GPT5 -mini	DeepSeek v3.1	GLM 4.5	Llama 3.1	Qwen 3
Glucose RMSE ^a	31.17	31.27	31.30	31.48	31.49	31.64
Calories MAE	52.71	56.00	56.57	59.44	66.76	65.00
Calories RMSE	87.14	84.11	89.35	91.01	100.06	100.70
Carb MAE	6.50	6.60	7.49	8.14	9.00	8.88
Carb RMSE	12.98	12.85	14.45	15.33	16.11	16.45
Fat MAE	3.44	4.69	4.19	4.18	4.61	4.48
Fat RMSE	6.37	7.44	7.45	7.11	7.86	7.58
Fiber MAE	0.96	1.15	0.94	1.24	1.22	1.28
Fiber RMSE	2.53	2.79	2.56	3.20	2.66	2.82
Protein MAE	3.11	3.15	3.13	3.07	4.00	3.25
Protein RMSE	5.09	5.34	5.31	4.94	6.41	5.03

^a. Mean Glucose RMSE was based on overall and postprandial blood glucose forecasting tasks. The filled color indicates the ranking of the value within the row. Carb: the abbreviation of carbohydrates.

IV. DISCUSSION

Our experiments demonstrate that augmenting continuous glucose forecasting with LLM-based nutrient estimations yields improvements in short-term glucose forecasting. Notably, the RMSE of postprandial glucose forecasting at 3

hours was reduced by up to 1.2 mg/dL, with the lowest 3-hour RMSE reaching 26.66 mg/dL, which remains competitive compared to the state-of-the-art CGM pretraining baseline (2-hour RMSE = 26.88 mg/dL) [25].

The correlation analysis provides mechanistic insight into why certain LLMs produce larger forecasting gains. We found the strongest associations between forecasting improvement and LLM estimation RMSE for carbohydrates and fat. This aligns with the physiology of glucose metabolism, where postprandial blood glucose is primarily influenced by carbohydrate intake, and fat consumption can mitigate the rate of glucose increase [26].

These results have practical implications for deploying LLM-assisted diet estimation in glucose-aware systems for diabetes self-management. First, the integration of a database with the LLM enhances the precision of glucose forecasting. Second, the novel RAG approach might benefit nutrient estimation compared to methods without RAG. However, the marginal improvement in glucose forecasting achieved with RAG may be limited, suggesting a need for cost-benefit analysis in practical applications. Third, the LLM selection for estimation should prioritize accuracy on both carbohydrate and fat estimation.

We also note several limitations. The Shanghai cohorts (T1DM $n = 12$, T2DM $n = 100$) reflect real-world, free-living recordings but remain limited in sample size and dietary variety. Dietary logs collected in free living conditions are subject to omissions, portion-size misestimation, and reporting bias; such upstream noise constrains the ceiling of any estimation method. The nutritional database used for evaluating LLM estimation is itself an imperfect proxy but not a “ground-truth” — recipes, cooking methods, and portion heterogeneity introduce unavoidable discrepancy. The proposed feature fusion strategy might not yet fully exploit patient-level information. The advantage of LLM-based nutrient estimation diminishes when models are trained on the combined multi-source features. Finally, our exploration of RAG is relatively simple, and advanced RAG technologies may bring greater improvements[27].

For future work, we recommend three directions. First, employ multimodal signals (meal images, timestamps, and portion scales) together with LLM text parsing to reduce ambiguity and improve nutrient estimates [28]. Second, refine the feature fusion framework to better preserve individual-level distinctions and achieve truly personalized, precision-oriented digital twins [29]. Third, expanding the dataset to incorporate data from more diverse populations and dietary cultures is crucial for assessing the generalizability and robustness of the proposed method.

V. CONCLUSION

This work validates the hypothesis that LLM-derived complete macronutrient and GI information significantly upgrades short-term glucose forecasting in free-living diabetes data. We proposed an LLM-augmented dietary estimation framework and integrated it into a CGM forecasting model. The experiments demonstrate the feasibility and effectiveness of incorporating enriched nutritional features into glucose forecasting. Future efforts will integrate multimodal meal inputs, patient-specific metabolic priors, and culturally diverse datasets to pursue a truly personalized glucose digital twin.

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